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9

Bioenergy

9.1 Learning Outcomes

After actively engaging with the material in this chapter, you should be able to

1. Discuss the main types of biomass fuels;
2. Explain how biomass resources store energy from the sun;
3. Explain with examples the meaning of primary and secondary biomass fuels;
4. Define the terms energy density, biomass yield, and carbon-neutral energy;
5. Calculate the yield of a biomass crop from a given area;
6. Calculate the energy content of a given quantity of biomass fuel;
7. Discuss advantages and disadvantages of biomass as a source of sustainable energy;
8. Explain the need for processing biomass materials;
9. Calculate the cost per unit of electricity from a biomass power plant;
10. Use available literature to critically contribute to the debate around using land for energy crops instead of food.

9.2 Overview

This chapter presents an introduction to bioenergy as a source of renewable energy. It defines and explains important terms and concepts relating to bioenergy and biomass fuels. A discussion of primary and secondary biomass materials and their use as feedstock for bioenergy production is presented. Important parameters relevant to the production of bioenergy, such as the yield and energy density of biomass materials, are explained with numerical examples. Advantages and disadvantages of bioenergy are also discussed and the processes needed to convert biomass materials into suitable forms for energy production are outlined. Finally, several problems to check and consolidate understanding of the material are provided at the end of the chapter.

9.3 Introduction

Carbon-rich substances, such as sugarcane, maize, and straw are good stores of chemical energy; hence, such substances are also known as carbon-based fuels. Energy stored in these substances can be released during combustion (i.e., burning). During combustion, these substances take in oxygen (O_2) from the atmosphere and release carbon dioxide (CO_2) and heat energy. Carbon-based fuels are of two types: fossil fuels and biofuels. Fossil fuels such as coal and oil are derived from carbon-rich matter that has taken millions of years to form, while biofuels are derived from organic matter that has been recently grown or formed and are known as biomass. Biomass is the name given for all living, or recently living, matter originating from plants and animals that represent a source of energy. The term bioenergy is used to refer to fuels derived from biomass. It includes solid fuels such as wood, liquid fuels such as biodiesel, and biogases such as bioethanol. Bioenergy is currently used in transport, heating, cooking, and for biopower (i.e., generating electricity). Examples of biomass include forestry products such as wood, agricultural by-products such as straw, industrial waste such as woodchips and wood off-cuts, animals, and humans waste, and plants grown specifically to be used as biofuels such as miscanthus, sugarcane, and maize. Traditional biomass (e.g., the burning of wood) is the oldest source of energy known to humans and it has been used since humans learned how to start a fire. In fact, traditional biomass is still a significant source of energy in many developing countries where wood is still used for heating, cooking, and lighting. For example, in some remote areas of some developing countries, horse manure is still being collected, dried, and used as fuel in special clay ovens for cooking. This is in addition to using it as a fertilizer.

Bioenergy is the fourth largest source of energy after oil, natural gas, and coal and it accounted for approximately 10% of the total global demand of

primary energy in 2015 [1]. Moreover, bioenergy represented the largest share of about 70% among all renewable energy sources providing about 57 EJ. This is about 10% of the world's total primary energy supply in 2016 [2]. In addition, biomass feedstock was the third largest renewable energy source for electricity generation in 2016 [3].

Biomass represents all living things on Earth that is contained in a thin surface layer of the earth known as the biosphere. The mass of the bioshpere is a very tiny fraction of the mass of the earth, yet it represents a huge store of chemical energy that is received from the sun, and much of it is available for us to harvest. Estimating the global potential of biomass energy and utilization is a very complex task due to the numerous forms, locations, and types of biomass feedstock [4]. However, some research estimates that the rate of energy storage by the total land biomass is about 2400 EJ per year, which is about four times the current global demand of primary energy [5]. Another study consisting of 19 different assessments concluded that energy crops alone should be able to provide about 1,300 EJ per year in 2050 [6].

It is important, however, to appreciate that although the biosphere represents a tiny fraction of the mass of the earth, its existence is so vital for life on this planet. If all plants were to disappear from the face of the earth, there would be no gases, such as oxygen and nitrogen, which are essential to sustain life.

9.4 Biomass as a Renewable Energy Source

A source of energy can only be considered renewable if it is sustainable or can be replenished in relatively short periods of time and its use does not lead to any significant emissions of greenhouse gases. Well, as we will soon discover, biomass fits, or at least it can be made to fit, this definition. Biomass is almost a carbon-neutral source of energy (i.e., the net release of carbon dioxide in the atmosphere) and is negligible when used responsibly for energy production. This is best illustrated with a comparison between the methods of extracting energy from a fossil fuel such as coal, and from a biomass source, such as a tree. Millions of years ago plants such as trees lived on the surface of Earth and while living, they absorbed carbon dioxide from the atmosphere and energy from the sun for several years. When they died, decayed, and compressed deep underground, they stored some of this energy and carbon dioxide and turned into coal. When coal is burned, this energy and carbon dioxide that has been stored for millions of years is released. Clearly, since it took millions of years to form coal, and therefore it is not a sustainable source of energy. In fact, coal is expected to be depleted during the next 30 to 80 years along with natural gas and oil [7]. On the other hand, if we plant a tree and leave it for several years

to grow, say 5 to 10 years, not millions, during this time it will be absorbing carbon dioxide from the atmosphere and energy from the sun and storing them. When the same tree is cut, it is processed to convert it into suitable form for use as a source of energy (e.g., wood logs or wood pellets). Later, when the tree is burned, it can only release the carbon dioxide it absorbed from the atmosphere while it was alive along with the stored energy. But if we plant another tree, the new tree will absorb a similar amount of carbon dioxide as that released by burning the first tree. Hence, if we plant a new tree for every tree we burn, the process should be carbon-neutral. Otherwise (i.e. if we burn more trees than we plant) this carbon neutrality will not hold and more carbon dioxide will be released than absorbed, which will lead to deforestation. However, carbon dioxide will also be released during processing and transporting of trees, and therefore, we must plant more trees than we use for energy in order to offset this extra carbon dioxide emission. Hence, fundamental to keeping this source of energy sustainable and carbon-neutral is a well-managed strategy for plantation and cultivation of trees.

9.5 Biomass as a Solar Energy Source

Biomass is a manifestation of solar energy that is stored in carbon containing compounds in biomass materials. Plants store energy from the sun during the process of photosynthesis. During this process, plants take in carbon dioxide and water from the atmosphere and energy from the sun, which is what makes plants biomass sources of fuel. During the process of photosynthesis, six carbon dioxide molecules react with six water molecules and with input energy from the sun according to the chemical equation

$$6\,CO_2 + 6\,H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6\,O_2 \tag{9.1}$$

The results of this reaction are glucose ($C_6H_{12}O_6$), which is a carbohydrate, and oxygen, which is released into the atmosphere. This is a constant cycle during which a plant absorbs water, carbon dioxide, and solar energy from the surroundings and releases oxygen into the atmosphere. This cycle is broken when a plant is burned, taking in oxygen and releasing carbon dioxide and heat energy. It is important to appreciate that carbon is a key element in biomass and is present in many compounds that make up the bulk of all living tissues. Examples of carbon-rich compounds found in biomass include glucose, cellulose, hemicellulose, lignin, lipids, and proteins. Compounds that contain only carbon, hydrogen, and oxygen are collectively known as carbohydrates and are the prime stores of energy in biomass.

9.6 Biomass as a Fuel

Extraction of energy from biomass normally requires combustion, which involves taking in oxygen and releasing heat energy and carbon dioxide. As an example, consider the combustion of methane (CH_4), which can be produced from biomass. A methane molecule consists of one carbon atom and four hydrogen atoms and each oxygen molecule consists of two oxygen atoms (O_2). During combustion reaction, each methane molecule reacts with two oxygen molecules according to the chemical equation

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + \text{Energy} \tag{9.2}$$

On the input (left-hand) side of the equation energy is stored as chemical energy in the fuel (i.e., in the methane) and oxygen, and on the output side (the right hand) of the equation, energy is released in the form of heat energy plus chemical energy in the carbon dioxide and water. The difference between the input and output energies define the energy or heat content, also known as the energy value of the fuel, in this case methane. When burning a fuel, if we know the chemical formula of the fuel and the relative atomic masses of the chemical elements of its molecules, we can calculate the amount of carbon dioxide emitted when a given amount of fuel is burned. For example, during the combustion of methane (CH_4), the relative atomic masses of the atoms of carbon and oxygen are 12 times and 16 times the mass of a hydrogen atom, respectively. This can be represented in the balanced chemical equation as

$$\begin{array}{ccccc} CH_4 & + & 2O_2 & \rightarrow & CO_2 + 2H_2O \\ 12+(4\times1)+ & & 2\times(2\times16) & \rightarrow & 12+(2\times16)+2\times(2\times1+16) \end{array} \tag{9.3}$$

That is, burning 16 tonnes of methane releases 44 tonnes of carbon dioxide. In other words, burning one tonne of methane produces 2.75 (=44/16) tonnes of carbon dioxide. The difference in energy between the input and output sides of the equation is about 55 GJ per tonne. Hence, we say that the energy content of methane is 55 GJ per tonne or 55 MJ/kg. In comparison, coal has an energy content of 24 GJ/tonne and is one of the highest producers of carbon dioxide because its chemical composition is largely carbon. Another common name for the energy content of fuel is the energy density, which we can now define as the amount of energy released when burning one kilogram of a given substance and is usually expressed in MJ/kg or in kWh/kg. The energy density for some fuels is provided in Table 9.1 [8, 9].

Table 9.1
Energy Density of Some Fuels

Fuel	Energy Density by Mass (GJ/t)	Energy Density by Volume (GJ/m^3)
Wood (oven-dried)	18	9
Coal (domestic)	28	25
Miscanthus	13	2
Maize grain (air-dried)	19	14
Wood (air-dried)	15	9
Domestic waste	9	1.5
Wood chips (30% MC)	12.5	3.1
Domestic heating oil	43	36
Baled straw	15	1.5
Sugarcane residue	17	10
Paper (stacked newspaper)	17	9

Example 9.1

Calculate the volume of airdried wood that is needed to bring one liter of pure water from 25°C to the boiling point (i.e., to 100°C). Given that the specific heat of water = 4,200 J /kg K, the mass of 1 liter of water is 1 kg, the energy density of airdried wood is 15 MJ/ kg, and the density of airdried wood is 600 kg/ m^{-3}.

Solution:

$$\text{Heat energy needed} = \text{mass} \times \text{temperature rise} \times \text{specific heat capacity}$$

Hence,

$$\text{heat energy needed} = 1\ \text{kg} \times (100\text{-}25)\ ^{o}\text{C} \times 4200\ \text{J/kgC}$$
$$= 315\ \text{kJ}$$

The energy released by burning 1 kg of wood is 15 MJ, but we only need 315 kJ, so the required mass of wood is $\frac{315\ \text{kJ}}{15000\ \text{kJ/kg}} = 0.021\ \text{kg}$

$$\text{Volume} = \frac{\text{mass}}{\text{density}}$$

and the volume of the required wood is

$$V = \frac{0.021\ \text{kg}}{600\ \text{kg/m}^3} = 3.5 \times 10^{-5}\ \text{m}^3 \quad \text{or } 35\ \text{cm}^3$$

Example 9.2

Repeat Example 9.1 above using miscanthus instead of airdried wood, given that its energy density by volume is 2 GJ/m^3 and the density of water is 1,000 kg/m^3 and explain your answer. Note that 1 liter of water is 0.001 cubic meter.

9.7 Primary Biomass Energy Sources

In this section, we will briefly consider various primary biomass sources of energy. Primary sources refer to plants that have been specifically grown to be used as biofuels. This means that a primary source requires a site that has been set aside for growing the biomass source. We will consider woody biomass derived from woody plants, cellulosic, sugary, and oilseeds crops in addition to microorganisms such as microalgae, which are a class of their own [8]. The key element in all of these is carbon, which represents the energy store.

9.7.1 Woody Biomass

The main source of woody biomass are trees that consist of mainly carbohydrates, lignin, cellulose, and hemicellulose. Woody biomass come from trees that have been planted specifically for energy production in specially designated forests. Well-managed planation and harvesting of trees for energy can provide a sustainable fuel source with reduced carbon dioxide emissions into the atmosphere.

Example 9.3

Use data in Table 9.1 to answer this question. The volume of one tonne of a certain grade of coal is 0.53 m^3. Calculate the volume of miscanthus that would be required to supply the same amount of energy as one tonne of coal.

Solution:

From Table 9.1, the energy content per tonne of coal is 28 GJ/t and the energy content per m^3 of miscanthus is 2 GJ/m^3; hence, to obtain 28 GJ from miscanthus we need

$$28 \div 2 = 14\text{m}^3$$

Note that 14 cubic meters of miscanthus are required in order to match the energy provided by just over half a cubic meter of coal. This low energy density of biomass can be a problem as it means higher costs of both storage and transport.

Example 9.4

Repeat Example 9.3 above for airdried wood instead of miscanthus. Explain your answer.

9.7.2 Cellulosic Materials

Cellulosic bioenergy crop is a main constituent of straw and many other major food crops. The most commonly known cellulose crop is miscanthus, also known as elephant grass. Miscanthus is flowery plant that originated in Asia and Africa and is currently grown in Europe and the United States. It has a low mineral content and high biomass yield and can be harvested annually, which makes it a favorite choice as a biofuel, outperforming maize and many other alternatives. The annual yield of miscanthus can be as much as 18 tonne per hectare with an energy density of 17 MJ/kg. However, these figures can vary considerably depending on factors such as location, climate, and nature of soil.

9.7.3 Sugary Crops

Sugar crops such as sugar beet, maize, and sugarcane can be grown to provide feedstock for energy production. The main biomass energy constituent of these is either sugar or starch, which when fermented produces bioethanol fuel. Sugarcane grown specifically for energy can be genetically modified to increase its fiber content. This results in significant increase in its biomass yield [10]. The high biomass yield of sugarcane and its high bioethanol productivity are major factors that contribute to the balance of greenhouse gas emission [11]. The residue of processing sugar crops for energy can be used as animal feed, which is an added advantage. In some locations, the yield of sugarcane can be as much as 98 tonne per hectare with an energy content of 16 MJ/kg [12]. Sugarcane is a significant crop for both energy and food, and it is estimated that 2.2 million hectares worldwide are used for its production. In Brazil for example, sugarcane annual business amounts to more than $20 billion [12].

9.7.4 Oilseed Crops

Oilseed crops such as soya beans, rapeseed, and sunflower seeds are grown primarily for the oil in their seeds, which has high levels of energy-rich fats as well as proteins. They have been grown for over thousands of years as food products. More recently, interest has been growing in these oily crops not only due to

their health benefits, but also for their use as feedstock to produce bioenergy. It is estimated that the global production, for example, of vegetable oils in 2006 was around 110 million tonnes [13]. Moreover, the oil from all these crops has many uses in industries such as cosmetics. The extraction of the oil from the seeds of these crops is by pressing, which a simple and inexpensive technique. The main bioenergy that is currently obtained from these oils is biodiesel, and the residue left from the oil extraction process can be used as animal feed. Typical yield of oily crops is approximately 8 to 15 tonnes per hectare per year [8].

9.7.5 Microalgae

Microalgae, found naturally in fresh waters that are currently farmed for energy production, have a high level of energy-rich fats and thus can be used as a bioenergy source, for example to produce biodiesel and biopetroleum. There are several attractive features that make microalgae good source of sustainable energy, such as the fact that they grow in water; hence, they don't have to compete for land that may otherwise be used for food production. There are also claims that they clean water of pollutants and help absorb carbon dioxide [8].

Grossly estimated yields of different annual biofuels per hectare are given in Table 9.2 [8]. It must be noted that these figures are only for guidance as actual figures depend on many factors such as the moisture content (MC), location, nature of soil, and climate. In fact, MC is a very important factor in determining the yield of a biomass and it is the norm that the yield is quoted in terms of dry tonnes with 0% moisture (i.e., oven-dried biomass).

9.8 Secondary Biomass Sources

A secondary biomass source refers to a source resulting from nonenergy use of biomass but has the potential for energy production. In other words, this is a biomass source that comes from wastes or refuse, such as industrial wastes, agricultural wastes, by-products, household refuse, and unwanted products of human activities. Examples of secondary sources include straw, which is an ag-

Table 9.2
Typical Approximate Yield of Dry Matter in Tonne Per Hectare Per Year

Fuel category	Examples	Yield of Dry Matter (t/ha.yr)
Woody biomass	Trees, hardwood	10 to 20
Cellulosic biomass	Miscanthus, seaweeds	10 to 60
Starch/sugary crops	Maize, sugarcane	10 to 35
Oily crops	Oilseeds (e.g., rape, sunflower)	8 to 15

ricultural by-product, wood off-cuts, rice husk, and human and animal wastes. Note that industrial waste, such as plastics, that have been synthesized from fossil fuels are not considered sources of renewable energy [8].

9.8.1 Wood Residues

It is estimated that the timber industry leaves behind about 15% of tree crops as wood waste on site, which can be utilized for energy production. Such wood residue may be used as firewood for power and heat or for making papers. In some countries like Sweden, these residues have been used for many years at an annual increasing rate of 10% [8].

9.8.2 Crop Coproducts

It is estimated that the annual global energy content of the residue form wheat and maize, the two major cereal crops, amount to 15 to 20 EJ. Similarly, the total energy content of the annual residues of sugarcane and rice is estimated at about 18 EJ. Increased reclamation of wastes and increased efficiency of conversion to electrical energy could result in up to 50 GW of generating capacity from the global sugar industry alone [8].

9.8.3 Animal Wastes

Cattle deposit their manure in the fields where they graze, which later decomposes aerobically through the respiration of a range of organisms that feed on it. This process takes in oxygen and releases carbon dioxide in the atmosphere, which is essentially a carbon-neutral process. However, wet slurry that is stored in bulk decomposes anaerobically and consequently releases methane and nitrous oxide, which can be dangerous to humans and the environment. However, under well-managed and controlled anaerobic digestion, biogas can be generated for energy use. Another common animal waste is poultry litter, which has relatively low moisture content allowing direct combustion with an energy content of 9 to 15 GJ per tonne [8].

9.8.4 Municipal Solid Waste

It is estimated that the average energy content of household waste, which is collected as municipal solid waste (MSW), is about 9 GJ per tonne. This everyday waste, which includes items such as packaging, newspapers, unwanted food, and clothing, represents a significant source of energy, while simultaneously helps keep the environment clean and reduces the emission of harmful gases into the atmosphere. In the industrialized countries, the average household generates about 1 tonne of MSW per year and in the United Kingdom about

20 million tonnes of food waste is generated each year [8]. In 2013, American households generated 254 million tonnes of MSW and recycled about 30% of it [14]. The United Kingdom recycled about 46% of the MSW generated in 2017 [15]. It is estimated that between 30%–60% of MSW is recycled in most western European countries and it is estimated that 24 waste-to-energy plants operating in England were treating 4 million tonnes of MSW in 2012 [16]. In the United States, 30 million tonnes of MSW generated about 14 billion kWh of electricity in 2016 [2]. The energy in the MSW is normally recovered using the process of incineration with heat recovery. The heat may be used directly for heating, usually district heating, indirectly for electrical power production, or for both combined heat and power generation. The estimated installed global capacity of power generation using MSW is about 3 GW, and about half of it is in Europe [8].

9.8.5 Commercial and Industrial Wastes

Commercial and industrial (C&I) wastes of organic origin come from a wide range of commercial activities, such as manufacturing, agricultural, and services industries. In 2016, the United Kingdom generated about 40.1 million tonnes of C&I wastes, of which 31.7 million tonnes were generated in England, and for 2017, this figure (i.e., for England) climbed up to 38 million tonnes. C&I wastes from the food industry, such as fats either in the form of used cooking oils or unwanted fatty tissue removed during meat processing, are potentially suitable either for direct combustion or conversion to biodiesel. Another important example of C&I wastes is worn tires; for example in the United Kingdom, it is estimated that 50 million tires are discarded each year with an energy content of 15 GJ per tonne, representing a valuable source of energy [8].

Example 9.5

It is claimed that if 10 million square kilometers of world land area is devoted to growing energy crops, it could contribute to annual bioenergy of about 250 EJ, which is approximately equivalent to 50% of the current global primary energy consumption.

a. Estimate the average annual bioenergy output per hectare that would be required to provide 250 EJ per year from the land area suggested above.
b. Given that the average energy density for maize and sugarcane is 17 GJ per tonne and their average annual yields are 35 and 15 tonnes per hectare respectively, calculate the annual energy yield of each in GJ per hectare. Compare your results with the claim made at the start of the question.

Solution:
10 million square kilometres is

$$1.0\times10^{7}\ \text{km}^2 = 1.0\times10^{7}\times10^{6}\,\text{m}^2 = 1\times10^{13}\,\text{m}^2$$

One hectare (ha) is 10,000 m^2; therefore, in hectares (ha) this is

$$\frac{1\times10^{13}\,\text{m}^2}{10000} = 1\times10^{9}\ \text{(ha)}$$

Therefore, the average bioenergy output per hectare per year is

$$\frac{250\times10^{18}\ \text{J}}{1\times10^{9}\ \text{(ha)}} = 2.50\times10^{11}\ \text{J/ha.yr}$$

In GJ this is

$$\frac{2.5\times10^{11}}{10^{9}} = 250\ \ \text{GJ/ha.yr}$$

For sugarcane the average annual yield is 35 tonnes per hectare; therefore the annual energy yield is

$$35\times17 = 595\ \ \text{GJ}$$

This is nearly 2.4 times the figure of 250 GJ in part (a) above.
For maize the annual yield is

$$15\times17 = 255\ \ \text{GJ/ha}$$

This is just 2% more than the claimed value of 250 GJ from part (a).

Example 9.6

Repeat Example 9.5 above using a grade of miscanthus with an annual yield of 12 tonnes per hectare and energy density of 10 GJ per tonne. Explain your results.

9.9 Advantages of Biomass

As the race continues for finding an alternative for fossil fuels, interest, and investment in biomass has been accelerating due to many of its advantages, which we discuss next.

9.9.1 Renewable

If well-managed, many sources of biomass, such as trees, maize, and sugarcane, can be readily replenished after use, which makes biomass truly renewable sources of energy. With replanting after responsible harvesting, plants can be maintained as a lasting source of energy.

9.9.2 Carbon-Neutral

When plants are used as fuel, only carbon dioxide released into the atmosphere is what was taken in during their life cycle. This carbon will be reclaimed from the atmosphere when these used plants are replenished. That is, this process of burning and replanting new plants is carbon-neutral and this is an exceptional feature of biomass.

9.9.3 Versatile

Biomass sources come in many different physical forms and can be converted into a variety of different forms of fuels. For example, they can be converted to liquid biofuels to power different types of machineries, or gaseous fuels for cooking, heating, and electricity generation.

9.9.4 Reduction of Waste

Recycling of household refuse, human and animal wastes, and commercial and industrial wastes provide, useful energy and also helps keep the environment clean and reduces the emissions of harmful greenhouse gases into the atmosphere.

9.9.5 Local

The local aspect is an important feature of biomass, which means that energy can be generated locally and cheaply. For example, a farmer can readily set up a small plant that uses the manure of cattle to produce methane gas, which can then be used for heating and cooking.

9.9.6 Abundant

Biomass sources in one form or another can be found or can be made available almost anywhere on Earth. This is because many biomass sources are integral part of the natural life cycle of our planet. However, to make sure that this remains sustainable, it must be carefully managed.

9.10 Disadvantages of Biomass

There are some claimed disadvantages to using biomass for energy production, which we discuss next.

9.10.1 Not Completely Clean

Even though biomass is a mostly carbon-neutral source of energy, the burning of wood still produces other emissions and smoke that pollutes the local environment, although the effects are much less than those produced by burning fossil fuels. However, compared to other renewables, such as wind and tidal, these effects are more evident.

9.10.2 Risk of Deforestation

Biomass fuels are sustainable sources of energy only if their resources are well-managed and maintained, otherwise the result will be widespread deforestation. Moreover, there is the argument that cutting down trees deprives wildlife of their natural habitat, which may lead to their extinction.

9.10.3 Requires Large Space

Planting crops for energy requires relatively large amounts of space, which may not always be available; for example, in built-up areas. This may mean having to plant biomass far away from the intended location of use and or processing, which results in added costs of transport. This makes them less favorable compared to other renewables such as solar energy. There is also the debate about using land for energy rather than for food production, which may lead to increased food prices.

9.10.4 Inefficient

Biofuels are not as efficient as fossil fuels. Fossil fuels such as diesel and petroleum are usually added to their biofuels counterparts to improve efficiency. This reduces the effectiveness of biofuels in cutting down dependence on fossil fuels.

9.10.5 Requires Water

One clear disadvantage of biomass is the large amount of water energy crops need. This can be costly especially if the source of water is remote and/or scarce. Moreover, this may result in water supply becoming less available for wildlife and/or humans. There is also the debate that water may also be used as a source of renewable energy, which is cleaner than biomass.

9.11 Processing of Biomass

Biomass fuel resources exit in a variety of physical forms (e.g., plants wood offcuts, liquids, and solids). Although in some cases they can be burned directly to release their stored energy; for example burning wood wastes in households, in most cases and certainly at large scales, biomass fuels require some processing before they can be used. This processing might be necessary for many reasons; for example, to make them more user-environment-friendly, easier for transportation, or make it easier to release their energy. The processing of biomass fuels can involve one or more of physical, thermochemical, or biochemical processes [8]. Examples of physical processing are the cutting and chipping of wood and the pressing of oily seeds. Cutting wood into smaller pieces makes it easier to transport and to use. Pressing of oilseeds is necessary to extract their oil, which is the end-user product.

Thermochemical processing involves the use of heat and possibly the use of some chemical reagents to convert biomass into more useful forms. The output from such processes may be heat, intermediate gaseous, or liquid fuels. An example of such a process is the combustion of solid biomass fuels in special ovens or stoves to provide heat. Another example is gasification in which a gaseous biofuel fuel is produced from solid biomass, such as woodchips under relatively high temperature in a carefully controlled environment that typically contains oxygen. The gas produced is called synthesis gas or syngas, which consists primarily of hydrogen, carbon monoxide, and carbon dioxide. Syngas is then cleaned refined and liquefied for use as a renewable energy fuel [8].

Biochemical processes almost always involve physical and chemical processing in addition to the fundamental biological process. These processes depend on microorganisms to convert biomass fuel into more useful and useable form of biofuel. A common example of biochemical processing is anaerobic digestion. This process uses bacteria to break down organic material into various organic acids, which after further decomposition, produces biogas that consists primarily of methane and carbon dioxide. Examples of the feedstock used are dung, sewage, food wastes and or refuse, and agricultural residue. The process takes place in a digester where digestion can be either wet or dry. In a wet digester, the feedstock (e.g., dung) is converted to a slurry by mixing it

with water. This requires a relatively large amount of water, which is clearly a disadvantage. In a dry digestion process, the moisture content in the digester is reduced, which means more energy input is required to mix the material; however, it alleviates the need for large quantities of water. Another example of biochemical processes is the process of fermentation. This is also an anaerobic process that uses yeast as the microorganism and sugar as the biomass fuel to produce ethanol.

For a detailed and well-cited treatment on the processing of biomass, the reader is referred to [8].

9.12 Summary

Biomass is organic material that comes from plants and animals in a variety of physical forms and varying levels of energy contents. Biomass is a renewable source of energy that stores significant amounts of solar energy. Biomass plants absorb the sun's energy during the process of photosynthesis. When biomass is burned, the chemical energy stored in it is released as heat. Some biomass may be burned directly to generate heat and/or steam to turn turbines to generate electricity or converted to liquid biogas that can be used as fuel for generating mechanical energy.

Examples of biomass:

- Woody plants and wood processing wastes;
- Agricultural crops and waste materials;
- MSW and household refuse;
- Animal manure and human sewage.

Biomass is abundant, environmentally friendly, and sustainable if well-managed and has a good potential as a major source of renewable energy to reduce dependence on fossil fuels. However, it can take up large areas of land and requires large amounts of water and if not well-managed can lead to deforestation.

9.13 Problems

P9.1 The chemical equation showing the combustion of ethanol is

$$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3\,H_2O + \text{Heat energy}$$

Calculate the amount of carbon dioxide generated when one tonne of ethanol is burned.

P9.2 In a given location, the annual yield per hectare of ovendried wood (i.e., with 0% moisture content) is 20 tonnes. Given that the energy density of ovendried wood is 18 GJ per tonne, calculate the amount of annual energy in joules and in kWh that can be generated by an area of one million square kilometers when used to produce ovendry wood.

P9.3 Use data from Table 9.1 to answer this question. The volume of one tonne of a certain grade of coal is 0.53 m^3. Calculate the volume of each of the following: stacked newspapers, domestic refuse, and maize, biofuels that would be required to supply the same amount of energy as one tonne of this coal, and explain your answers.

P9.4 A land of 10 square kilometers is to be dedicated to biofuel energy crops. There is a choice of miscanthus or maize. Given that the average energy content for miscanthus and maize is 13 and 15 GJ per tonne respectively and that their average annual yields are 12 and 19 tonnes per hectare, respectively, calculate the average annual bioenergy output in kWh for each choice of biofuel.

P9.5 An area of 20,000 hectares is used to grow miscanthus to provide district heating. If miscanthus has a yield of 12 tonnes per hectare per year and an energy content of 18 GJ per tonne:

a. Calculate the annual energy supply represented by this area.

b. Assuming a household needs on average 3,000 kWh a year for heating and that 30% of the energy produced by the miscanthus is lost during processing and delivery, how many households could be heated by this area?

P9.6 A 40-MW straw-fired power plant uses 250,000 tonnes of straw per year and its average annual electricity output is 300 GWh. Given that the energy density of straw is 15 GJ per tonne, calculate the average annual efficiency of straw to electricity conversion and the average capacity factor of the plant. What are the main disadvantages of using straw as a fuel for a power station?

P9.7 The council of a small town of 18,000 households is considering the possibility of an energy from waste (EfW) power generation plant to mitigate the cost of landfills for its annual 20,000 tonnes of domestic refuse. Assuming that the energy density of household refuse is 10 GJ per tonne:

a. Calculate the number of annual kilowatt-hours the proposed plant would generate, using all the local household refuse and running at a waste-to-energy conversion efficiency of 30%.

b. Using the result of (a) above, determine a suitable rating in kW for the generator assuming the plant will be running with a capacity factor of 82%;

c. Given that the average household consumption of electricity is 3,200 kWh per year, what fraction of the total load demand would be provided by the plant?

P9.8 Explain why biomass is, in principle, a solar source of energy.

P9.9 Explain what is meant by the following terms: bioenergy, primary and secondary biomass, and carbon-neutral energy.

P9.10 Explain what makes biomass a source of energy and what makes it a sustainable source of energy.

P9.11 Write a concise essay why it is important to develop large-scale systems to produce biofuels for motor vehicles.

P9.12 Write a short essay of about 500 words in which you give your own critical views on the question of using land to produce energy crops.

P9.13 Discuss a few ways to reduce your own household refuse.

P9.14 Discuss two advantages and two disadvantages of biomass as source of energy.

P9.15 What are the shortcomings of using straw as the main fuel for a power generation plant?

P9.16 Recently, there has been a growing global interest in the production of liquid biofuels from purpose-grown energy crops. Using appropriate literature, write an appropriately referenced paper of about 800 to 1,000 words that includes the following:

1. A description of two types of primary energy source plants outlining briefly the methods used to extract the biofuels from them.

2. A discussion of possible benefits of a major shift from fossil-based liquid fuels to liquid biofuels and the main obstacles, and critically evaluate the debate for and against the use of land for energy rather than for food.

3. An assessment of the current use and future potential of liquid biofuel.

P9.17 Use the literature to write an appropriately referenced essay of about 1,000 words assessing and critically reviewing the current and future potential of using liquid biofuels in the transportation industry.

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